MOSPEC

SWITCHMODE SERIES NPN POWER TRANSISTORS

... designed for use in high-voltage,high-speed,power switching in inductive circuit, and switchmode applications such as switching regulator's,converters.

NPN

2SC2625

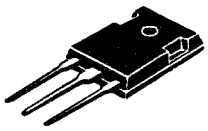
10 AMPERE
SILICON POWER
TRANASISTORS
400 VOLTS
80 WATTS

FEATURES:

*Collector-Emitter Sustaining Voltage-
$V_{CEO(sus)}$ = 400 V (Min)

* Collector-Emitter Saturation Voltage -
$V_{CE(sat)}$ = 1.2 V (Max.) @ I_C = 4.0 A, I_B = 0.8A

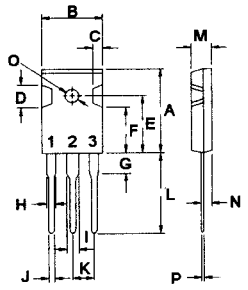
* Switching Time - t_f = 1.0 us (Max.) @ I_C =5.0 A



TO-247(3P)

MAXIMUM RATINGS

Characteristic	Symbol	2SC2625	Unit
Collector-Emitter Voltage	V_{CEO}	400	V
Collector-Base Voltage	V_{CBO}	450	V
Emitter-Base Voltage	V_{EBO}	7.0	V
Collector Current - Continuous - Peak	I_C I_{CM}	10 20	A
Base current	I_B	3.0	A
Total Power Dissipation @T_C = 25°C Derate above 25°C	P_D	80 0.64	W W/°C
Operating and Storage Junction Temperature Range	T_J , T_{STG}	-55 to +150	°C



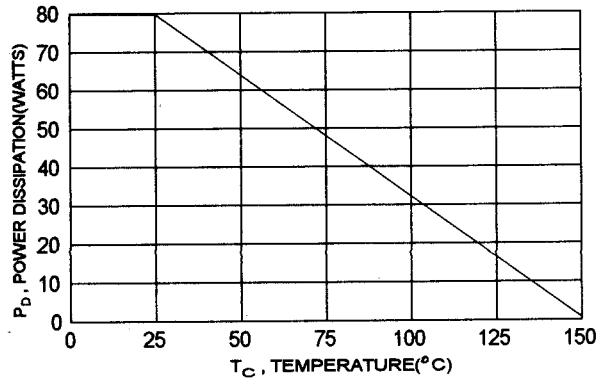
PIN 1.BASE
2.COLLECTOR
3.EMITTER

DIM	MILLIMETERS	
	MIN	MAX
A	20.63	22.38
B	15.38	16.20
C	1.90	2.70
D	5.10	6.10
E	14.81	15.22
F	11.72	12.84
G	4.20	4.50
H	1.82	2.46
I	2.92	3.23
J	0.89	1.53
K	5.26	5.66
L	18.50	21.50
M	4.68	5.36
N	2.40	2.80
O	3.25	3.65
P	0.55	0.70

THERMAL CHARACTERISTICS

Characteristic	Symbol	Max	Unit
Thermal Resistance Junction to Case	Rθjc	1.56	°C/W

FIGURE -1 POWER DERATING

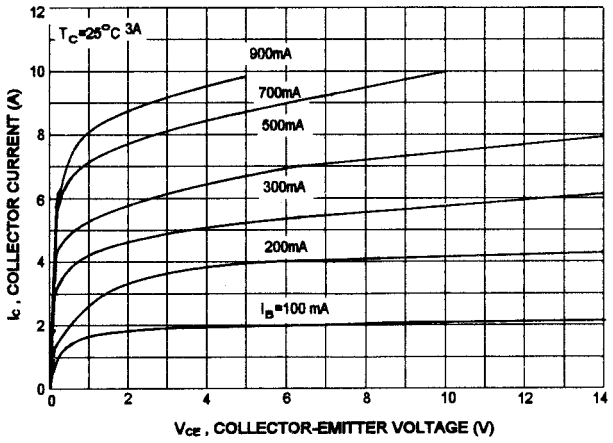


ELECTRICAL CHARACTERISTICS (T_C = 25°C unless otherwise noted)

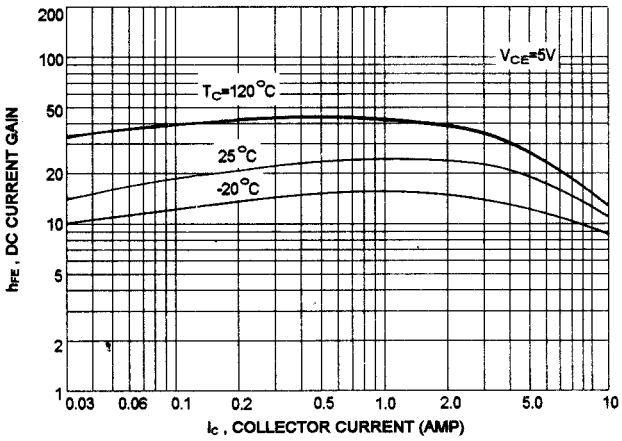
Characteristic	Symbol	Min	Max	Unit
OFF CHARACTERISTICS				
Collector-Emitter Breakdown Voltage (I_C= 10 mA, I_B=0)	$V_{(BR)CEO}$	400		V
Collector-Base Breakdown Voltage (I_C= 1.0 mA, I_E=0)	$V_{(BR)CBO}$	450		V
Collector Cutoff Current (V_{CB}= 450 V, I_E= 0)	I_{CBO}		1.0	mA
Emitter Cutoff Current (V_{EB}= 7.0 V, I_C= 0)	I_{EBO}		100	uA
ON CHARACTERISTICS (1)				
DC Current Gain (I_C= 4.0 A,V_{CE}= 5.0 V)	hFE	10		
Collector-Emitter Saturation Voltage (I_C= 4.0 A,I_B= 800 mA)	$V_{CE(sat)}$		1.2	V
Base-Emitter Saturation Voltage (I_C= 4.0 A,I_B= 800 mA)	$V_{BE(sat)}$		1.5	V
SWITCHING CHARACTERISTICS				
On Time	V_{CC}= 150 V, I_C= 5.0 A I_{B1}=-I_{B2}= 1.0 A R_L=30 ohm	t_{on}		1.0 us
Storage Time		t_s		2.5 us
Fall Time		t_f		1.0 us

(1) Pulse Test: Pulse Width =300 us,Duty Cycle ≦ 2.0%

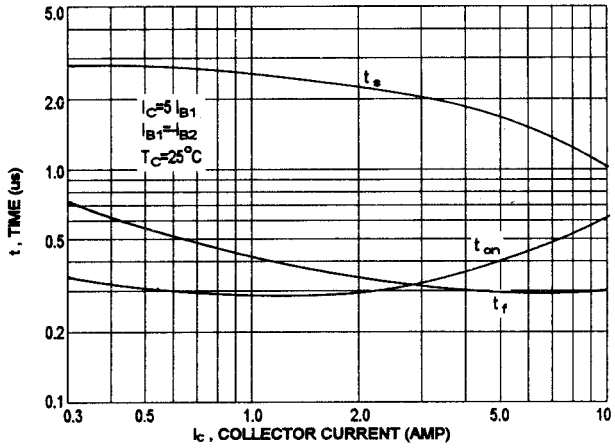
Ic - Vce



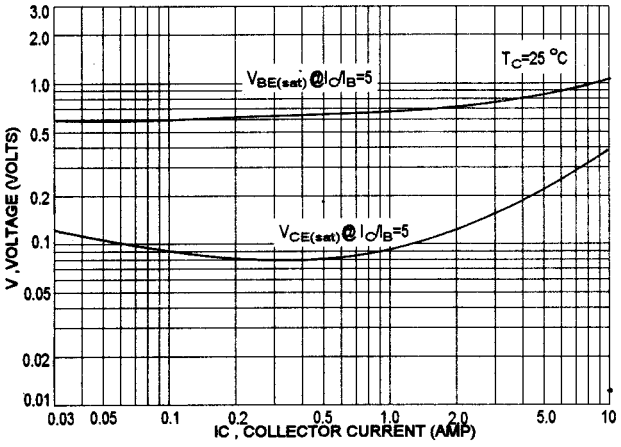
DC CURRENT GAIN



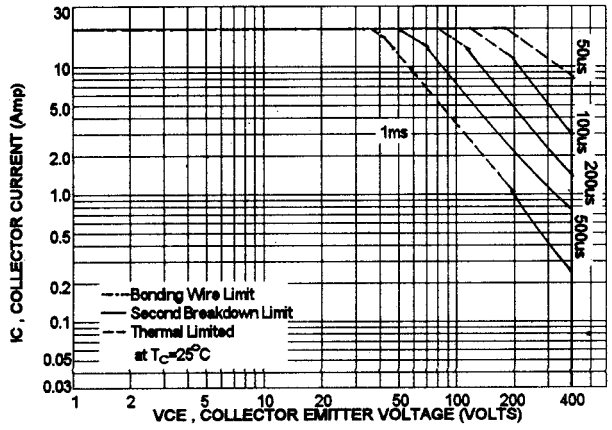
SWITCHING TIME



"ON" VOLTAGES



ACTIVE-REGION SAFE OPERATING AREA (SOA)



There are two limitation on the power handling ability of a transistor:average junction temperature and second breakdown safe operating area curves indicate I_C-V_{CE} limits of the transistor that must be observed for reliable operation i.e., the transistor must not be subjected to greater dissipation than curves indicate.

The data of SOA curve is base on $T_{J(PK)}$=150 °C;T_C is variable depending on conditions. second breakdown pulse limits are valid for duty cycles to 10% provided $T_{J(PK)}$≤150°C,At high case temperatures, thermal limita - tion will reduce the power that can be handled to values less than the limitations imposed by second breakdown.